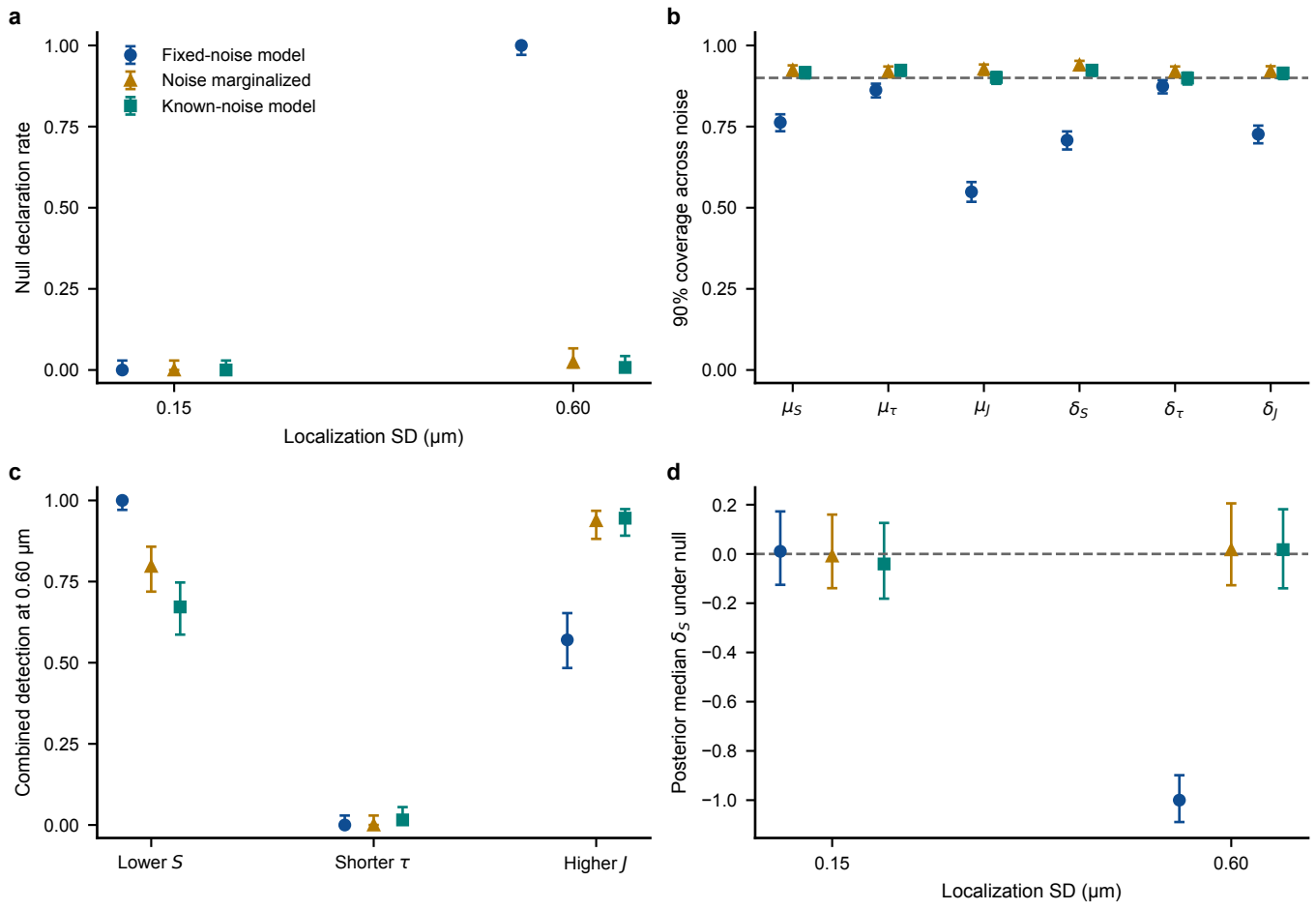
a
Fixed-noise model
Noise marginalized
Known-noise model
Null declaration rate
1.00
0.75
0.50
0.25
0.00
0.15
0.60
Localization SD (μm)
b
90% coverage across noise
1.00
0.75
0.50
0.25
0.00
μ_S
μ_τ
μ_J
δ_S
δ_τ
δ_J
c
Combined detection at 0.60 μm
1.00
0.75
0.50
0.25
0.00
Lower S
Shorter τ
Higher J
d
Posterior median δ_S under null
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0.15
0.60
Localization SD (μm)